Hannes Kaufmann

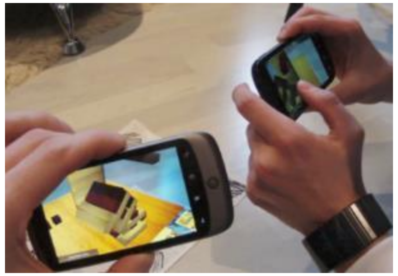
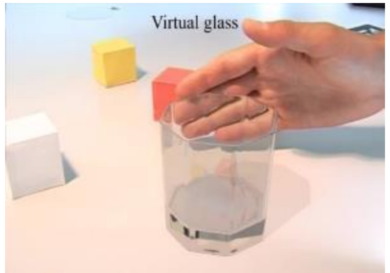
Multimedia

Vorbesprechung

Hannes Kaufmann

Research Unit Virtual and Augmented Reality

Institute of Visual Computing & Human-Centered Technology

TU WIEN: VIRTUAL & AUGMENTED REALITY

Univ.Prof. Mag. Dr.
Hannes Kaufmann

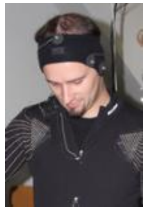
Ao.Univ.Prof. Dr.
Horst Eidenberger

Research Staff










Postdocs: Peter Kán, Iana Podkosova, Christian Schönauer, Francesco De Pace, Hugo Brument

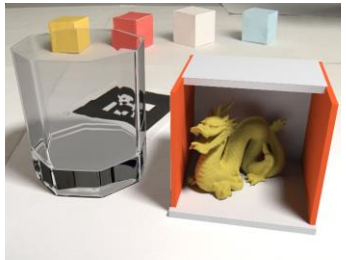
PhDs: Khrystyna Vasylevska, Emanuel Vonach, Soroosh Mortezapoor, Mohammad Ghazanfari, Matteo Bosco
+ 3 external PhDs

Students: 15 graduate and undergraduate students involved in research

Components of Mixed Reality

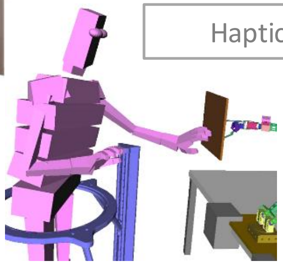
Total:
138,9
FPS

Real Time Ray Tracing in VR



High Quality Rendering in AR

Firefighter Training with Virtualizer



Haptic Feedback

Input

Rendering & Viewing

Localization (Tracking)

Interaction

Distribution

Collaboration

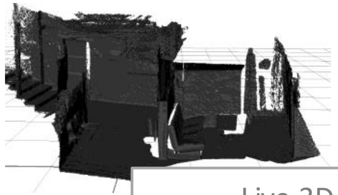
Authoring

Perception

Output

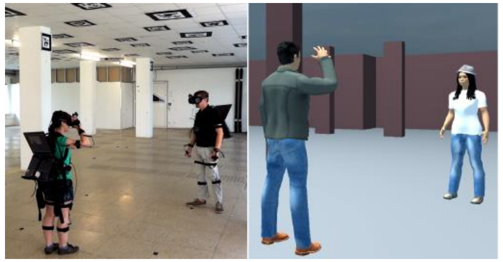
Wide Area Multi-User Optical Localization

ioTracker

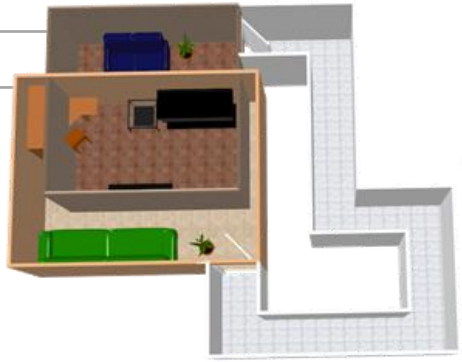


Live 3D Reconstruction for First Responders

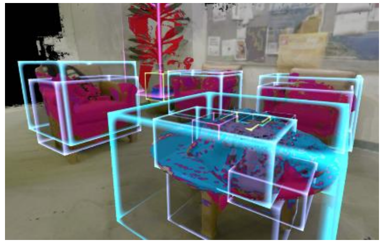
Shared Collaborative Virtual Spaces Navigation



Redirected Walking



Object Detection and Segmentation in Live 3D Reconstruction



Automated 3D Model Generation

Real Walking in Large VR - Platform Requirements

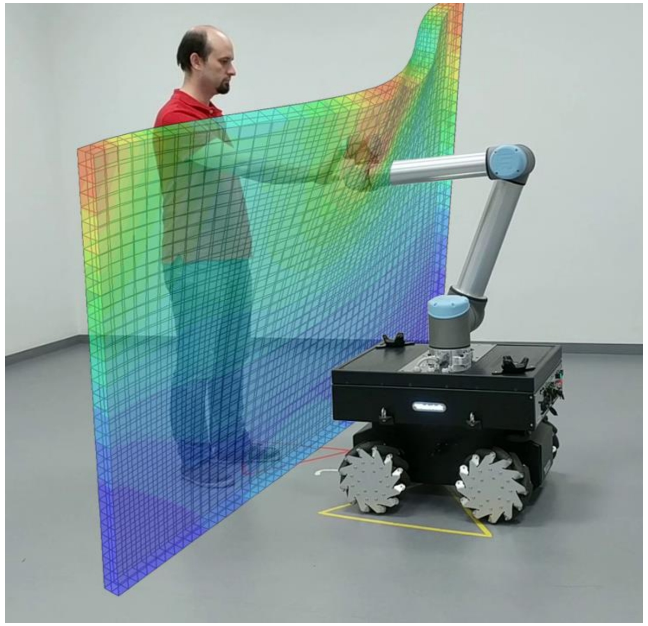
- Multi-user for **Gaming**
- Users navigate by walking
- **Low-cost** large area tracking system
- Full-body animated avatars
- Virtual elevator
- Haptic interaction
- 2 ways of using space
  - Collaboration
  - Non-shared space

ILLUSION WALK
IMMERSIVE EXPERIENCE PROVIDER

TU WIEN

Image credit: IllusionWalk KG

Interaction: Haptic Feedback in Large Scale VR



- Robot actuated physical props for passive haptic feedback

Challenges:

- Unexpected user movements
- Close direct interaction & multiple users in environment
- Haptic response from differently stiff materials required

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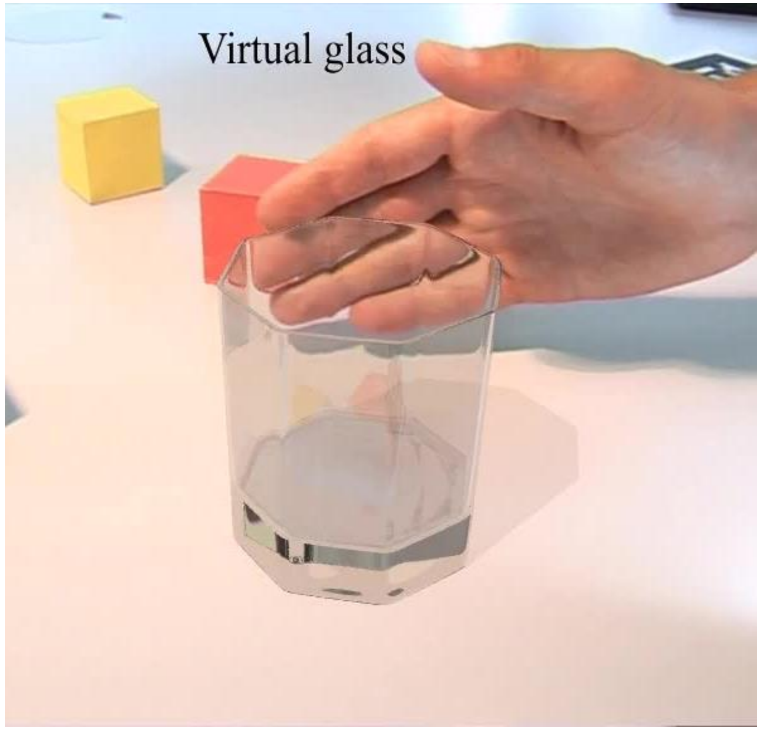
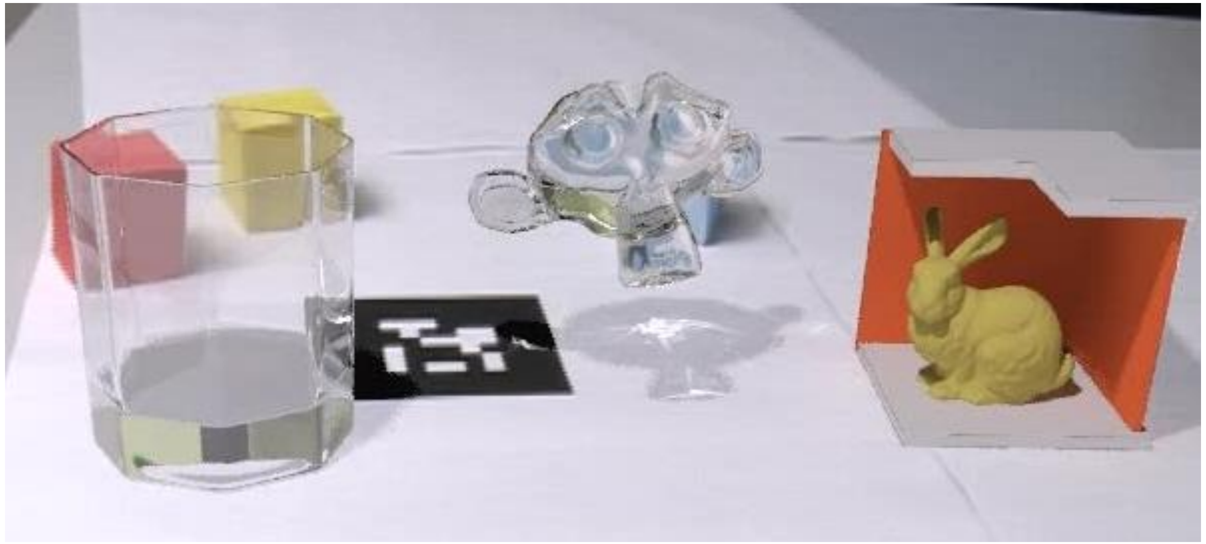
Real-time Ray Tracing in VR/AR

- Real-time ray tracing for AR environments
  - Correctly simulate light interaction between real and virtual objects
  - Photorealistic rendering in AR





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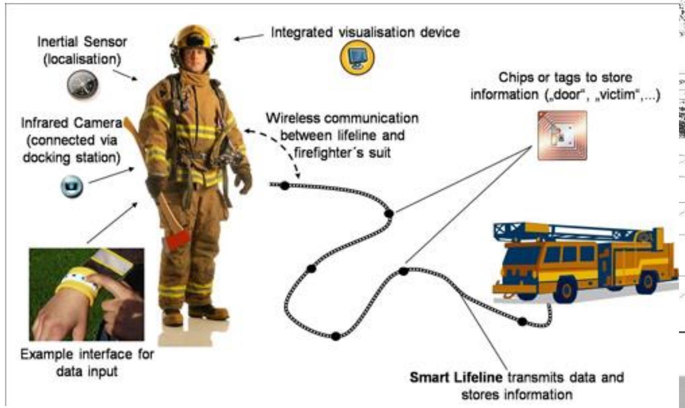
Applications: Medical & Biosensor

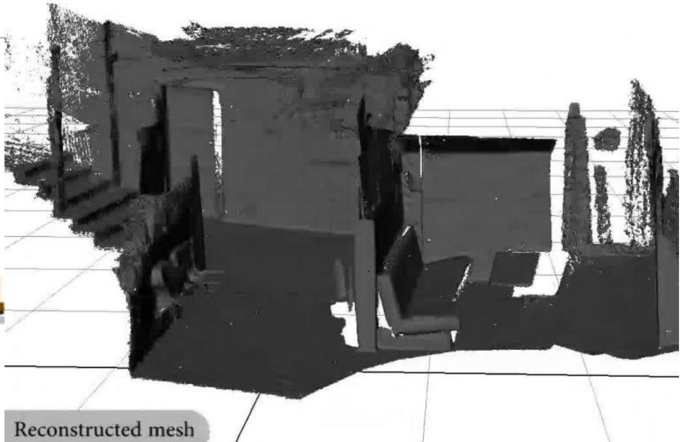
- PLAYMANCER

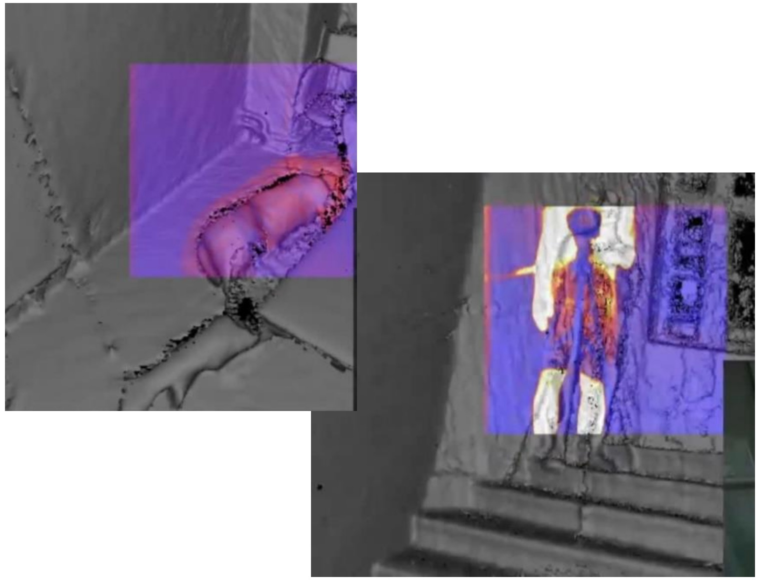
  3D-Serious Game Environment with real-time motion capturing and bio-signal feedback for physical rehabilitation

- ProFiTex

  Support fire fighters with mission-relevant information based on various sensor data



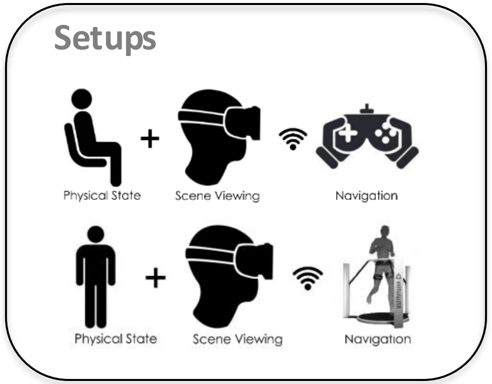


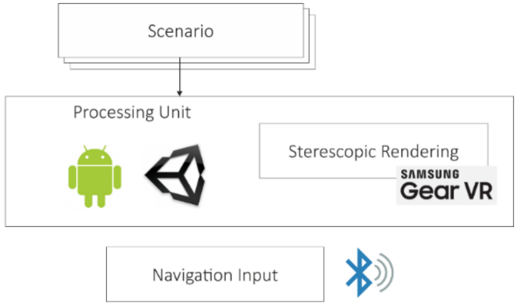


Applications: Training

Virtual Simulation and On-Site Training for First Responders

- Mobile immersive virtual reality training platform
  - On-site squad leaders of disaster relief units
- Save cost and time, increase training time





A. Mossel, M. Froeschl, C. Schoenauer, A. Peer, J. Goellner, H. Kaufmann (2017), VROnSite: Towards Immersive Training of First Responder Squad Leaders in Untethered Virtual Reality, *IEEE Virtual Reality, March 2017, Los Angeles, USA*

MasterMind Development GmbH FFG

Current Infrastructure @ TU



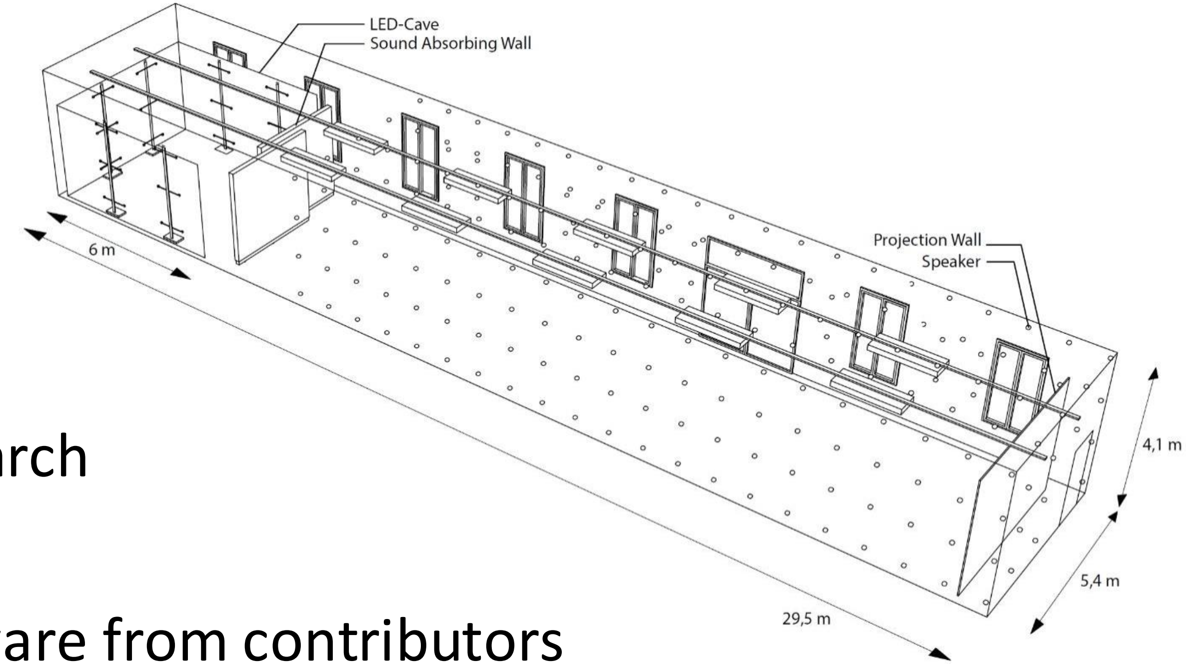

- VR Lab at Science Center (~140 m²)
  - Haptic research (SFB), large-scale VR
- Hallway Lab (~20 m²)
  - All teaching, student lab work & hand ins

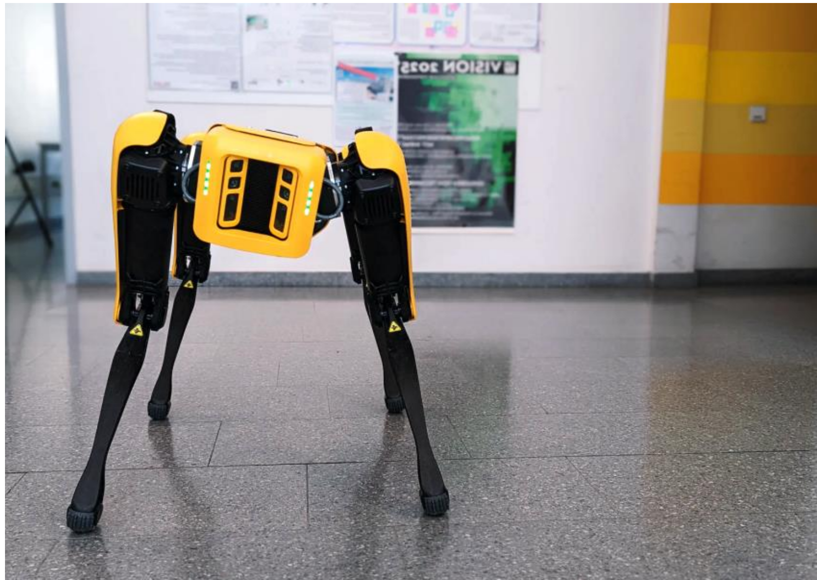

- VR Lab (~20 m²)




Infrastructure: TU Mixed Reality Lab

- TU Mixed Reality Lab (2026)
  - All 8 faculties contributing
  - 30x5x4m room (Fav.str. 9, EG)
  - Precise motion tracking in whole room
  - >50 HMDs (VR & AR) for teaching and research
  - LED Cave, LED Wall, Holographic displays
  - Hosting Spot robot, Lidar scanners & hardware from contributors
  - Splitting of room possible for multiple use-case scenarios
  - Lab manager (technician/senior scientist)





Organisational: Multimedia VO - Dates

- VU 4.0 SwS, 6.0 ECTS, LVA 193.148
- Tuesday 14:00-15:30 – starts on time! (s.t.)
- Begin: 10.3.2026 – End 19.5.2026

- In Presence in HS Fav 1

VO – Dates

Vorlesungstermine / Lecture Dates:

1. VO: 10.3. 14:00-16:00
2. VO: Inverted Classroom - Self Learning (Videos) please watch until 17.3. to understand the next lecture
3. VO: 17.3. 14:00-16:00

- Conference & Easter Break

4. VO: 14.4. 14:00-16:00
5. VO: 21.4. 14:00-16:00
6. VO: 28.4. 14:00-16:00
7. VO: 5.5. 14:00-16:00
8. VO: 12.5. 14:00-16:00
9. VO: Inverted Classroom - Self Learning (Videos)
10. VO: 19.5. 14:00-16:00
11. VO: Inverted Classroom - Self Learning (Videos)

Exam

- Written Exam (next 2 dates)
  - Tue 9.6.2026, 10:00 - 12:00, EI 7 HS (Gußhausstr.)
  - Tue 23.6.2026, 10:00 – 12:00, EI 7 HS (Gußhausstr.)
- Contact:
  - Tel.: 01/58801 18860
  - Email: Hannes.kaufmann@tuwien.ac.at
  - Favoritenstr. 9-11; 4. Stock; Stiege 3; HC 04 15

Grading

- Written exam for lecture part
- Grade of the practical part (group-work) results from quality of the hand-ins (defined by certain criteria) and the oral hand-in meeting.
- **To take the written exam, the practical part must be positive**.
- For a positive VU grade, the grade for the lecture part and for the practical part must be positive (>50% each).
- Final VU grade:
(66% lecture grade + 33% practical grade) / 2

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Content

- Multimedia Definition
- 4 Media Types:
  - Text
  - Audio: Psychoacoustics, Digital Audio, MIDI, Compression Standards (MP3) Spatial Audio (Binaural, Ambisonics)
  - Image: Color, HDR, Compression (e.g. JPEG)
  - Video: History (analog, TV), Compression (MPEG, H.264)
  - Other Human Senses
- Compression Standards
- Network Distribution/Streaming
- Examples

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Other Lectures offered by VR-Group

- Audio & Video Produktion (von Karyn Laudisi)
- Bachelorarbeit für Informatik und Wirtschaftsinformatik

Master:

- Virtual and Augmented Reality VO + LU, 2+4 ECTS (SS)
- Virtual Reality Advanced Topics; 188.456; 3 ECTS (in WS)
- Multimedia Interfaces VU (in WS)
- Praktikum
  - Praktikum aus Visual Computing 1 & 2
  - Projekt Medizinische Informatik
  - Project in Computer Science 1 & 2
- Diplomarbeiten

Questions?

VO – Videos and Slides online in TUWEL course, linked in TISS.